Fraud Detection API

Version 1.1.0 / OpenAPI 3.1.0

Use this reference to submit transactions, review alerts, and check model readiness, feature drift, and retraining status.

Interactive documentation http://localhost:8000/docs
OpenAPI schema http://localhost:8000/openapi.json

Endpoints

`GET /health`	Report API status and whether the model is loaded.
`GET /ready`	Check whether the XGBoost model is available.
`POST /score`	Score a transaction and create a case for REVIEW.
`GET /cases`	List alert cases and counts by status.
`PATCH /cases/{transaction_id}`	Update case status or feedback.
`GET /monitoring/drift`	Return the latest feature drift summary as JSON.
`GET /monitoring/drift/report`	Open the latest Evidently report as HTML.
`GET /monitoring/retraining`	Return the latest retraining status as JSON.

Readiness and monitoring

GET /health returns status and model_loaded. GET /ready returns HTTP 503 when the model is unavailable. The drift endpoints return HTTP 404 before a report is available.

GET /monitoring/retraining returns waiting_for_monitor until a monitoring run writes a status.

Score a transaction

POST /score / application/json

Transaction request

`transaction_id`	Required string, 1 to 128 characters
`TransactionAmt`	Required number, zero or greater
`TransactionDT`	Required integer, seconds relative to the dataset origin
`card1`	Optional integer card proxy
`addr1`	Optional integer address proxy
`DeviceInfo`	Optional string describing the device
`P_emaildomain`	Optional purchaser email domain string
`account_id`	Optional account identifier string
`card_id`	Optional card identifier string
`device_id`	Optional device identifier string
`extra_features`	Optional object with boolean, numeric, string or null values

Example request body

```
{
  "transaction_id": "demo-001",
  "TransactionAmt": 125.50,
  "TransactionDT": 14000000,
  "card1": 1000
}
```

FraudScore response

The required fields are transaction_id, fraud_probability, risk_score, decision, and model. The score is the fraud probability rounded to an integer on a 0-100 scale. At a probability of 0.20 or higher the decision is REVIEW. Otherwise it is ALLOW.

Optional fields include xgb_probability, gnn_score, ring_score, ring_id, and card_history. CardHistory contains an integer transaction_count and a numeric mean_amount.

HTTP 409 indicates a transaction ID already used with different data. HTTP 422 indicates invalid input. HTTP 503 indicates the scoring model is unavailable.

Review alert cases

GET /cases / PATCH /cases/{transaction_id}

List cases

Use the optional status query parameter to filter by open, investigating, or closed. Set limit to an integer from 1 to 250. The default is 100.

```
GET /cases?status=open&limit=25
```

The response contains counts by status and a cases array. Each case includes the score, status, label, created_at, and updated_at.

Update a case

Provide the transaction_id in the path. The CaseUpdate body accepts status and label. Set label to fraud or legitimate, or use null to clear it. A closed case requires a label.

```
PATCH /cases/demo-001
Content-Type: application/json

{"status": "closed", "label": "fraud"}
```

A successful update returns the updated case. HTTP 404 indicates the case was not found. HTTP 422 indicates an invalid request or a case update that violates a validation rule.

Validation errors

Request validation returns HTTP 422 with an HTTPValidationError object. Its detail array contains ValidationError objects with loc (a list of field names or indexes), msg (a message), and type (an error code). An error may also include input and ctx.